

Hall Ticket No. -

Code: R122131

**Stanley College of Engineering & Technology for Women (A)**

**B.E (CSE, CME & AI&DS) III Semester (Main) Examinations-January-2023**

**Mathematics-III (P&S)**

**Time: 03 Hours**

**Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.  
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.  
iii. Missing data, if any, may be suitably assumed.

**Part- A**

**6X2=12M**

1. a) State and Prove Addition theorem for two events. (2M CO1 BT1)
- b) If the mean of Binomial distribution is 3 and variance is  $\frac{9}{4}$ , Obtain the value of n. (2M CO2 BT1)
- c) Define Karl Pearson Coefficient Correlation, (2M CO3 BT1)
- d) Given  $y' = x^2 - y$ ,  $y(0)=1$ , find correct to four decimal places the value of  $y(0.1)$  by using Euler's method. (2M CO4 BT3)
- e) Are these vectors linearly dependent, verify  $(2,1,0), (1,2,5), (5,4,5)$  (2M CO5 BT2)
- f) Find the area A under the normal curve to the left of  $z = -1.78$  [Tab. Value: 0.4625] (2M CO2 BT1)

**Part-B**

**4X12=48M**

2. a) State and Prove Baye's Theorem. (6M CO1 BT2)
  - b) Box A contains 5 red and 3 white marbles and box B contains 2 red and 6 white marbles. If a marble is drawn from each box, what is the probability that they are both of same colour? (6M CO1 BT3)
3. a) Derive the mean and variance of Binomial Distribution. (6M CO2 BT3)
  - b) Fit a Poisson distribution for the following data and calculate the expected frequencies

|      |     |    |    |   |   |
|------|-----|----|----|---|---|
| x    | 0   | 1  | 2  | 3 | 4 |
| f(x) | 109 | 65 | 22 | 3 | 1 |

(2M CO2 BT3)

4. a) Fit a parabola to the following data

|   |    |    |   |    |    |
|---|----|----|---|----|----|
| x | 1  | 2  | 3 | 4  | 5  |
| y | 10 | 12 | 8 | 10 | 14 |

(2M CO3 BT3)

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- b) Two horses A and B were tested according to the time to run a particular track with the following results. Test whether the two horses have the same running capacity.

|         |    |    |    |    |    |    |    |
|---------|----|----|----|----|----|----|----|
| Horse A | 28 | 30 | 32 | 33 | 33 | 29 | 34 |
| Horse B | 29 | 30 | 30 | 24 | 27 | 29 | -- |

(Tab. Value=2.201)

(6MCO3 BT3)

5. a) Use Taylors series method to find the approximate value of  $y$  when  $x=0.1$ ,  $x=0.2$  given

that  $y(0)=1$ ,  $y'=3x+y^2$

(6M CO4 BT3)

- b) Use R-K method to evaluate  $y(0.1)$  and  $y(0.2)$  given that  $y' = x + y$ ,  $y(0) = 1$ .

(6M CO4 BT3)

6. a) Define Vector Space, Sub-Space and Give examples.

(6M CO5 BT1)

- b) Show that the vectors  $(1, 2, 3)$ ,  $(3, -2, 1)$ ,  $(1, -6, -5)$  form linearly dependent set and hence find its basis.

(6M CO5 BT4)

7. a) A random variable  $X$  has the following probability function

(6M CO1 BT3)

|        |     |      |      |      |      |       |
|--------|-----|------|------|------|------|-------|
| $X=x$  | 1   | 2    | 3    | 4    | 5    | 6     |
| $P(x)$ | $K$ | $3K$ | $5K$ | $7K$ | $9K$ | $11K$ |

Determine i)  $k$  ii) Expectation iii) Variance

- b) For the continuous probability function  $f(x) = kx^2 e^{-x}$  when  $x \geq 0$ , find

i)  $k$  ii) Mean iii) Variance

(6M CO1 BT3)

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**Stanley College of Engineering & Technology for Women (A)**

**B.E (CSE,CME&AI&DS) III Semester (Main) Examinations-January-2023**

**Discrete Mathematics**

**Time: 03 Hours**

**Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.  
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.  
iii. Missing data, if any, may be suitably assumed.

**Part-A**

**6X2=12M**

1. a. Summarize: (i) Conjunction (ii) Implication. [2M CO1 L2]  
b. If  $A = \{2, 3, 4, 5\}$  and  $B = \{0, 1, 2, 3\}$  find  $A \cap B$ . [2M CO4 L1]  
c. Define Scalar matrix with example. [2M CO2 L1]  
d. Find the characteristic roots of the matrix  $A = \begin{bmatrix} 5 & 3 \\ 3 & 4 \end{bmatrix}$ . [2M CO2 L1]  
e. Define: (i) Congruence relation (ii) Equivalence relation. [2M CO3 L1]  
f. Summarize (i) Bipartite graph (ii) Complete graph. [2M CO5 L2]

**Part-B**

**4X12=48M**

2. a) Explain (i) Tree traversal (ii) Quantifiers. [6M CO3 L2]  
b) Prove that  $(p \leftrightarrow q) = (p \rightarrow q) \wedge (q \rightarrow p)$ . [6M CO3 L5]
3. a) Define (i) Pigeon hole principle (ii) Permutation (iii) Combination. [6M CO2 L1]  
b) Simplify the boolean expression:  $A = XY + X(Y+Z) + Y(Y+Z)$ . [6M CO4 L4]
4. Illustrate Cayley-Hamilton theorem for matrix  $A = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}$  [12M CO5 L2]
5. Demonstrate  $H = \{0, 2, 4\}$  is subgroup of the group  $(G, +_6)$ , where  $G = \{0, 1, 2, 3, 4, 5\}$  [12M CO4 L2]
6. a) If  $U = \{2, 4, 6, 8, 10, 12, 14, 16\}$ ,  $A = \{2, 6, 10\}$  and  $B = \{4, 8, 10, 12, 14, 16\}$ , then find (i)  $A - B$  (ii)  $B - A$  (iii)  $A \cap B$  [6M CO3 L1]  
b) By using contradiction, prove that:  $\sqrt{2}$  is irrational. [6M CO4 L2]
7. A bag contains 10 red marbles, 10 white marbles, and 10 blue marbles. What is the minimum no. of marbles you have to choose randomly from the bag to ensure that we get 4 marbles of same color? [12M CO2 L2]

**Stanley College of Engineering & Technology for Women (A)****B.E (CME & AI&DS) III Semester (Main) Examinations-January-2023****Concepts in Computer Organization and Microprocessor****Time: 03 Hours****Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.  
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.  
iii. Missing data, if any, may be suitably assumed.

**Part-A****6X2=12M**

1. a. Design the block diagrams for programming in hardware and software approaches [2M CO1 BTL6]  
b. Define Micro operation. [2M CO2 BTL1]  
c. Illustrate phases of the instruction cycle. [2M CO3 BTL3]  
d. Define hit ratio. [2M CO1 BTL1]  
e. Compare microprocessor and microcontroller [2M CO3 BTL5]  
f. Classify the addressing modes of 8051 microcontroller [2M CO4 BTL4]

**Part-B****4X12=48M**

2. a) List and Explain the elements of bus design along with the timing of synchronous and asynchronous bus operations. [6M CO1 BTL1]  
b) Show arithmetic logic shift unit in detail. [6M CO2 BTL1]
3. a) Distinguish the different types of fixed point representation? Explain each with an example. [6M CO3 BTL4]  
b) Discuss floating point representation in detail. [6M CO3 BTL2]
4. a) Describe control memory in CPU. [6M CO2 BTL2]  
b) Summarize input-output and interrupt operation. [6M CO3 BTL2]
5. a) Construct zero, one, two and three address instruction formats in detail. [6M CO2 BTL2]  
b) Elaborate cache memory in detail. [6M CO2 BTL6]
6. a) List Arithmetic and Logical 8085 instructions with examples. [6M CO4 BTL1]  
b) What is DMA? Draw 8257 DMA controller. [6M CO5 BTL1]
7. a) Draw and explain 8051 architecture/block diagram. [6M CO4 BTL2]  
b) How does RS232 work? [6M CO2 BTL1]

# Stanley College of Engineering & Technology for Women (A)

## B.E (AI&DS) III Semester (Main) Examinations-January-2023

### Database Management Systems

Time: 03 Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.  
 ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.  
 iii. Missing data, if any, may be suitably assumed.

#### Part-A

5X2=10M

1. a. What is the role of DBA? [2M CO1 BTL1]
- b. Difference between view and table. [2M CO2 BTL2]
- c. Define specialization with an example. [2M CO3 BTL1]
- d. What is the Static Hashing? [2M CO4 BTL1]
- e. What are the ACID properties of transaction? [2M CO4 BTL1]
- f. Define lock based protocol. [2M CO5 BTL1]

#### Part-B

5X10=50M

2. a) Discuss briefly the architecture of a database system. [8M CO1 BTL2]
- b) Difference between file system and DBMS. [4M CO1 BTL2]
3. a) Explain the various operations in relational algebra with examples. [8M CO2 BTL2]
- b) Develop a function for largest of 3 numbers using PL/SQL. [4M CO2 BTL6]
4. a) What is normalization? Explain in detail about 1NF and 2NF. [6M CO3 BTL2]
- b) Explain functional dependency with examples. [6M CO3 BTL3]
5. a) What is an index ? Explain dense index with an example. [6M CO4 BTL3]
- b) Explain conflict serializability. [6M CO4 BTL3]
6. a) Analyze briefly lock based protocol in transactions and its types. [8M CO5 BTL3]
- b) Explain Buffer Management [4M CO5 BTL3]
7. a) What is Join operator in SQL? Explain different Joins in SQL with examples. [6M CO2 BTL3]
- b) Explain Time Stamp based protocol. [6M CO5 BTL3]

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## Stanley College of Engineering & Technology for Women (A)

B.E (CSE, CME & AI&DS) III Semester (Main) Examinations-January-2023

### OOPS Using JAVA

Time: 03 Hours

Max. Marks-60

- Note:**
- First Question is Compulsory. Answer any Four out of remaining Six questions.
  - Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
  - Missing data, if any, may be suitably assumed.

#### Part-A

6X2=12M

- a. Define Type Casting and give an example. [2M CO1 BTL1]
- b. What is overriding in Java? [2M CO2 BTL1]
- c. List out the built-in Exceptions in Java. [2M CO3 BTL1]
- d. What is a Wrapper Class? [2M CO4 BTL1]
- e. What are the sources of Event? [2M CO5 BTL1]
- f. What is a package? List out built-in packages of Java. [2M CO3 BTL1]

#### Part-B

4X12=48M

- a) What is a Constructor? Discuss types of constructor with the examples. [6M CO1 BTL2]
- b) Explain String Buffer and String Builder with an example program. [6M CO1 BTL2]
- a) How to design and implement an interface in Java? Give an example. [6M CO2 BTL5]
- b) Explain the process of defining and creating a package with suitable examples. [6M CO2 BTL3]
- a) Write a java program that illustrates the application of multiple catch statements. [6M CO3 BTL4]
- b) Differentiate multiprocessing from multithreading. Explain the different ways to create a thread in Java. [6M CO3 BTL2]
- a) Compare and contrast Array List and Linked list. Explain. [6M CO4 BTL2]
- b) Describe Iterator and List Iterator. Explain different ways to iterate over a list. [6M CO4 BTL5]
- a) Describe each of the mouse and keyboard events handling methods [6M CO5 BTL2]
- b) Explain MVC architecture with the diagram. [6M CO5 BTL4]
7. Write short notes on the following.
  - a) Final Keyword usage [4M CO1 BTL2]
  - b) Serialization [4M CO4 BTL2]
  - c) Lambda Expressions [4M CO5 BTL2]

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## Stanley College of Engineering and Technology for Women(A)

B.E(CSE , CME & AI&DS) II Semester (Backlog) Examinations January-2023

### Chemistry

Time: 3 hours

Max.Marks:60

#### PART-A

Note: Answer all questions(Compulsory)

5 X 2=10M

- 1 Why do we use  $\text{CaCO}_3$  as reference for calculating hardness of water? Mention its units.
- 2 Mention the role of salt bridge in electrochemical cell.
- 3 Define the terms i) Monomer ii) Functionality of polymer.
- 4 Define knocking and explain the significance of octane number.
- 5 Mention any two applications of carbon nanotubes.

2 M CO1 L2

2 M CO2 L4

2 M CO3 L1

2 M CO4 L1

2 M CO5 L3

#### PART-B

Note: Answer all questions

5 X 10= 50M

- 6 a. Illustrate water softening by ion – exchange process with the help of neat labeled diagram.
- b. Discuss the sacrificial anodic protection and impressed current cathodic protection.
- OR
- c. Define alkalinity of water. How is it determined?
- d. Discuss the various factors influencing rate of corrosion.
- 7 a. Calculate the emf of the following cell at  $25^\circ\text{C}$ :  $\text{Zn(s)}/\text{Zn}^{2+}(0.001\text{M})//\text{Ag}^+(0.0001\text{M})/\text{Ag(s)}$ . Given  $E_0$  of  $\text{Zn}^{2+}/\text{Zn} = -0.76\text{ V}$  and  $E_0$  of  $\text{Ag}^+/\text{Ag} = 0.80\text{ V}$ .
- b. Explain the construction and working of methanol-oxygen fuel cell.

5 M CO1 L3

5 M CO1 L2

5 M CO1 L2

5 M CO1 L3

5 M CO2 L3

5 M CO2 L2

#### OR

- c. Analyze thermodynamic parameters for cell reaction taking place in galvanic cell.
- d. Discuss the construction and cell reaction of Calomel Electrode.
- 8 a. Write a note on molecular orbital theory.
- b. Write the preparation, properties and uses of Polylactic acid.

5 M CO2 L4

5 M CO2 L3

5 M CO3 L3

5 M CO3 L2

OR

- c. Describe the energy level diagram of NO and specify its magnetic property 5 M CO3 L3
- d. Explain the preparation, properties and applications of i) PET and ii) BUNA-S 5 M CO3 L2

- 9 a. Write any four requirements of a good fuel and define HCV. 5 M CO4 L2
- b. Calculate the gross and net calorific value of a coal sample having the following composition; C=80%, H=7%, O=3%, S=3.5%, N=2.1% and ash=4%. 5 M CO4 L4

OR

- c. Explain the ultimate analysis of coal to ascertain its quality and its significance 5 M CO4 L3
- d. Discuss any three important fractions of crude oil. 5 M CO4 L2

- 10 a. Explain the twelve principles of green chemistry. 5 M CO5 L3
- b. Write a short note on i) nanowires and ii) nanocrystals. 5 M CO5 L2

OR

- c. Illustrate transesterification with a suitable example. 5 M CO5 L3
- d. Explain the classification of composites based on their matrix. 5 M CO5 L2

**B.E (CSE,CME & AI&DS) II Semester (Backlog) Examinations January-2023**  
**Environmental Science**

**Time: 3 hours**

**Max.Marks:60**

**PART-A**

**5 X 2=10M**

**Note: Answer all questions (Compulsory)**

- |                                                                               |    |
|-------------------------------------------------------------------------------|----|
| 1 List some major conflicts in sharing water resources between two countries. | 2M |
| 2 Differentiate renewable and non renewable energy resources.                 | 2M |
| 3 Justify the statement "Ecosystem regulates itself".                         | 2M |
| 4 Define Photochemical Smog.                                                  | 2M |
| 5 Differentiate between a national park and sanctuary.                        | 2M |

**PART-B**

**5 X 10= 50M**

**Note: Answer all questions**

- |                                                         |    |
|---------------------------------------------------------|----|
| 6 a. What are dams? What are the benefits and problems? | 5M |
| b. Discuss about soil erosion and desertification.      | 5M |

**OR**

- |                                                                                                                                      |    |
|--------------------------------------------------------------------------------------------------------------------------------------|----|
| c. Explain the effects of Fertilizer and Pesticide related problems.                                                                 | 5M |
| d. Explain the environmental issues related to fossil fuel and nuclear power energy generation.                                      | 5M |
| 7 a. What are Ecological pyramids? Explain why some of these pyramids are upright while others are inverted in different ecosystems. | 5M |
| b. Explain about types of ecosystems.                                                                                                | 5M |

**OR**

- |                                                                                                                                             |    |
|---------------------------------------------------------------------------------------------------------------------------------------------|----|
| c. Define an ecosystem with the help of pond ecosystem, explain the different components of the ecosystem and their functions in ecosystem. | 5M |
| d. Define and classify ecological succession.                                                                                               | 5M |
| 8 a. Discuss the value of Biodiversity and threats to Biodiversity.                                                                         | 5M |
| b. Explain about bio graphical classification of India.                                                                                     | 5M |

**OR**

- |                                                                                 |    |
|---------------------------------------------------------------------------------|----|
| c. Write about Endemic and Endangered species of India.                         | 5M |
| d. Define hotspots of biodiversity. Explain about hotspots in India.            | 5M |
| 9 a. What is Soil Pollution? Mention the causes, effects and remedial measures. | 5M |
| b. Write short notes on forest and wild life acts.                              | 5M |

**OR**

- |                                                                             |    |
|-----------------------------------------------------------------------------|----|
| c. Describe about solid waste management.                                   | 5M |
| d. Discuss the salient features of Air Act in detail.                       | 5M |
| 10 a. How can one conserve water in a community level and individual level. | 5M |
| b. Explain about Disaster Management Cycle.                                 | 5M |

**OR**

- |                                                                                   |    |
|-----------------------------------------------------------------------------------|----|
| c. Describe about infrastructure and development of India.                        | 5M |
| d. What is global warming? Explain about the responsible parameters for the same. | 5M |

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## Stanley College of Engineering and Technology for Women(A)

B.E(CSE , CME , AI&amp;DS) II Semester (Backlog) Examinations January-2023

## Basic Electrical and Electronic Circuits

Time: 3 hours

Max.Marks:60

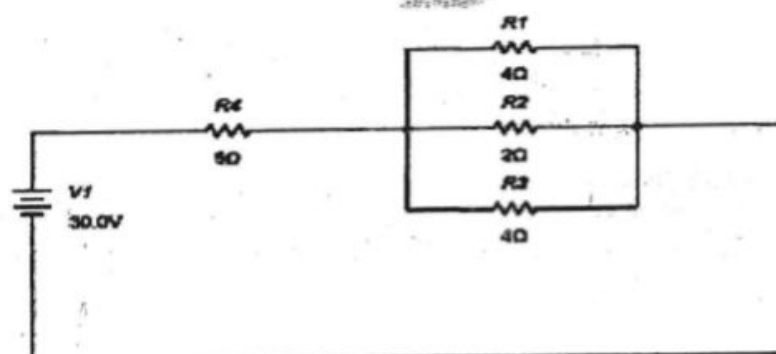
## PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

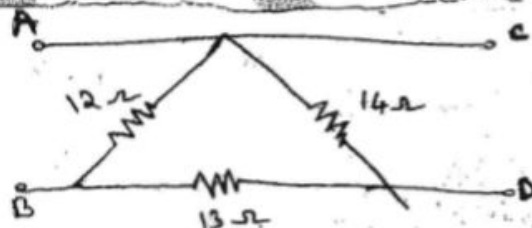
1. Determine the total current in the circuit shown

2



2. Obtain star connected equivalent circuit for the delta connected circuit shown.

2



3. Find the total current flowing through a germanium diode at room temperature when voltage across the diode is 0.2V and reverse saturation current is 20mA.

2

4. List any two advantages of h-parameter model of BJT.

2

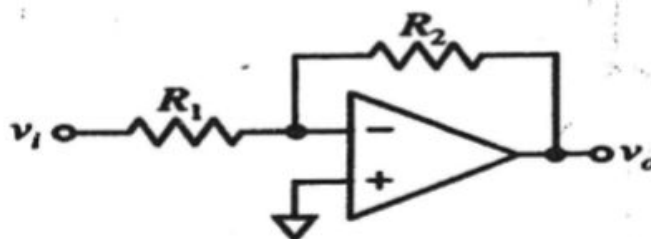
5. For the circuit shown below:

2

$$R_1 = 27 \text{ k}\Omega$$

$$R_2 = 39 \text{ k}\Omega$$

Assuming the OpAmp is ideal and input voltage is  $v_i = 4 \text{ V}$ , what is the output voltage  $v_o$ ?



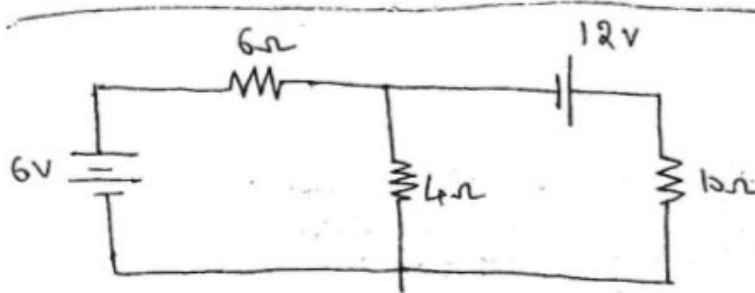
## PART-B

Note: Answer all questions

5 X 10 = 50M

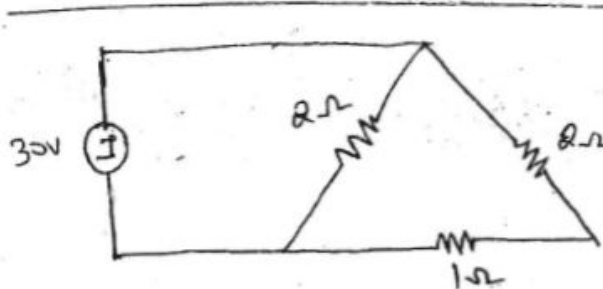
6. a. What is the voltage across  $4\ \Omega$  resistor in the circuit shown below?

5



- b. Determine the current delivered by the source in the circuit shown below.

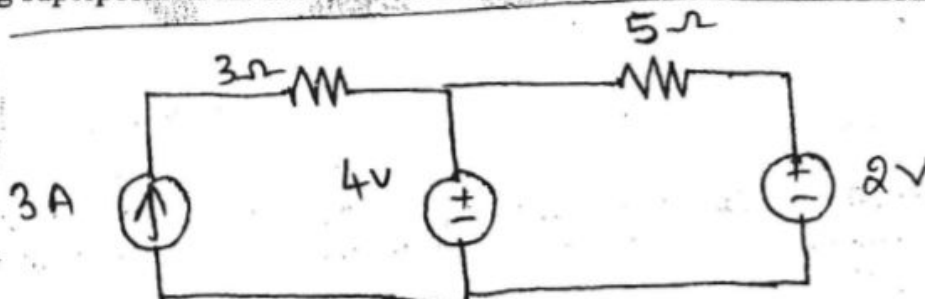
5



OR

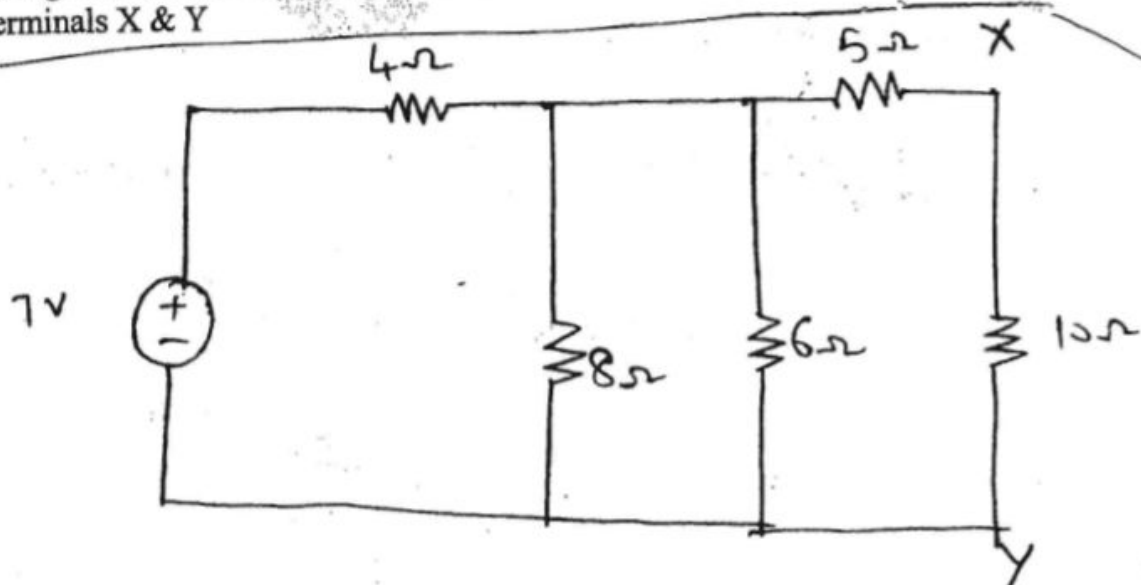
- c. Using superposition theorem, determine the current through the  $5\ \Omega$  resistor.

5



- d. Using Thevenin's theorem, determine the Thevenin's equivalent circuit across the terminals X & Y

5



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**Stanley College of Engineering and Technology for Women (A)**  
**B.E (CSE, CME, AI&DS & IT) II Semester (Backlog) Examinations January-2023**  
**Data Structures with C**

Time: 3 hours

Max.Marks:60

**Part-A**  
**Note: Answer all questions (Compulsory)**

**5 X 2 = 10M**

1. List the examples for linear and non linear data structures?
2. List the applications of stack?
3. What is sparse matrix and give its purpose?
4. Write the recursive procedure to perform pre-order traversal of a Binary Tree?
5. Define binary search tree?

2MCO8 L1

2MCO8 L1

2MCO7 L1

2MCO2 L2

2MCO9 L1

**Part-B**

**Note: Answer all questions**

**5 X 10 = 50M**

6. a) Explain binary search algorithm. Compute its time complexity.

OR

10M CO1, 6 L2

- c) Distinguish between quick sort and heap sort with a suitable example.

10M CO4 L2

7. a) Explain the procedure to evaluate postfix expression. Evaluate the following postfix expression  $7\ 3\ 4\ +\ -\ 2\ 4\ 5\ /\ +\ * \ 6\ /\ 7\ +\ ?$

OR

10M CO8 L2

- c) Explain the operations performed on simple queue with an example.

10M CO8 L2

8. a) Write an algorithm to delete an element anywhere from doubly linked list.

OR

10M CO7 L1

- c) Write applications of single linked list to represent polynomial expressions.

10M CO7 L1

9. a) What traversals can be performed on binary trees and how binary tree can be represented using Array and Linked list.

10M CO9 L1

OR

- c) Illustrate breadth first search graph traversal algorithm with a suitable example.

10M CO9 L4

10. a) Write an algorithm for insert and delete an element from AVL Tree. Explain with suitable examples.

10M CO7 L1

OR

- c) Demonstrate various resolution techniques in Hash table with suitable examples.

10M CO9 L3

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**Stanley College of Engineering and Technology for Women (A)**

**B.E (CSE/EEE/ECE/IT/CME/AI&DS) II Semester (Backlog) Examinations January-2023**

**Mathematics--II**

Time: 3 hours

Max.Marks:60

**PART-A**

**Note: Answer all questions (Compulsory)**

**5X2=10M**

- 1 Solve  $(x^2 + y^2) dx + 2xy dy = 0$  2 M L3
- 2 Find the C.F. of  $(D^2 + D + 1)y = 0$  2 M L2
- 3 Define Beta function and Gamma function. 2 M L1
- 4 Find the Eigen Values of the Matrix  $A = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$  2 M L3
- 5 Find  $L\{t^2 \sin at\}$  2 M L3

**PART-B**

**Note: Answer all questions**

**5X10=50M**

- 6 a. Solve  $\frac{dy}{dx} + y \cos x = y^3 \sin 2x$  5 M L3
- b. Find the Orthogonal trajectories of  $\frac{x^2}{a^2 + \alpha} + \frac{y^2}{b^2 + \alpha} = 1$ ,  $\alpha$  being the parameter. 5 M L3
- OR**
- c. Solve  $(1 + e^{xy}) dx + e^{xy} (1 - \frac{x}{y}) dy = 0$  5 M L3
- d. Solve  $p = \sin(y + xp)$ . Also find its singular solutions. 5 M L3
- 7 a. Solve  $(D^2 + 4)y = e^x + \sin 2x + \cos 2x$  5 M L3
- b. Solve  $y'' + y = \sec x$  by the Method of Variation of parameters. 5 M L3
- OR**
- c. Solve  $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + 2y = x \log x$  5 M L3
- d. Solve  $(D^2 - 4D + 3)y = 3e^x \cos 2x + 2xe^{3x}$  5 M L3
- 8 a. State and prove the relation between Beta and Gamma Function. 6 M L4
- b. Evaluate  $\int_0^2 x^3 \sqrt{2-x} dx$  using Beta And Gamma function. 4 M L5
- OR**
- c. State and Prove Rodrigue's Formula. 6 M L4
- d. Evaluate  $\int_0^a x^4 \sqrt{a^2 - x^2} dx$  4 M L5

- 9 a. Verify Cayley – Hamilton Theorem for the Matrix 6 M L4  
 $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$  and also find  $A^{-1}$

- b. Discuss the nature of the Q.F.  $x^2 + 4xy + 6xz + 2yz - y^2 + 4z^2$  4 M L4

OR

- c. Find Eigen Values and corresponding Eigen Vectors of 8 M L3

$$\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

- d. Using Cayley – Hamilton Theorem, find  $A^8$  if  $A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$  2 M L3

- 10 a. Find the Laplace Transform of (i)  $t e^{2t} \sin 3t$  (ii)  $e^{-t} \cos^2 t$  5 M L3

- b. Find the Inverse Laplace Transform of  $\frac{2s+12}{s^2+6s+12}$  and  $\frac{s}{(s+2)^2}$  5 M L3

OR

- c. Find  $L \{ t \cos at \}$  2 M L5

- d. Using Laplace Transform method, solve 8 M L5  
 $(D^2 + 1)y = 6 \cos 2t, t > 0$ , If  $y = 1, Dy = 1$  when  $t = 0$ .

## Stanley College of Engineering and Technology for Women (A)

B.E (CSE/EEE/ECE/IT/CME/AI&amp;DS) I Semester (Backlog) Examinations January-2023

## Mathematics-I

Time: 3 hours

Max.Marks:60

## PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

1. Test the convergence of the series  $\sum (1 + \frac{1}{n})^{-n^2}$  2M
2. Verify Cauchy's mean value theorem for  $f(x) = x^2$  and  $g(x) = x^3$  in  $[1, 2]$  2M
3. If  $u = \frac{x+y}{1-xy}$  and  $v = \tan^{-1} x + \tan^{-1} y$  then prove that  $u$  and  $v$  are functionally dependent. 2M
4. Find grad  $f$  if  $f = x^2y + y^2x + z^2$  at  $(1, 1, -2)$  2M
5. Find the first approximation of the root lying between 0 and 1 of the equation  $x^3 + 3x - 1 = 0$  by Newton-Raphson formula. 2M

## PART B

Note: Answer all questions

5 X 10 = 50M

6. (a) Discuss the convergence or divergence of  $\frac{x}{1.2} + \frac{x^2}{2.3} + \frac{x^3}{3.4} + \dots$  5M
- (b) Examine the convergence of the series  $1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \dots$  5M

(OR)

- (c) Test the convergence of the series  $1 + \frac{2}{5}x + \frac{6}{9}x^2 + \dots + \frac{2^n - 2}{2^{n+1}}x^{n-1} + \dots$  ( $x > 0$ ) 5M
- (d) Show that the exponential series  $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$  converges absolutely for all values of  $x$ . 5M

7. (a) Find the evolute of the parabola  $y^2 = 4ax$  5M

- (b) Verify Cauchy's Mean value theorem for the functions  $f(x)$  and  $f^1(x)$  in  $(1, e)$  given  $f(x) = \log x$  5M

Code: S222830

(c) Prove that  $\frac{\pi}{6} + \frac{1}{5\sqrt{3}} < \sin^{-1}\left(\frac{3}{5}\right) < \frac{\pi}{6} + \frac{1}{8}$  10M

8. (a) If  $u = x + y + z$ ,  $y + z = uv$ ,  $z = uvw$  then show that  $\frac{\partial(x,y,z)}{\partial(u,v,w)} = u^2v$  5M

(b) If  $u = x^2 + y^2$ ,  $x = at^2$ ,  $y = 2at$  then find  $\frac{du}{dt}$  5M

(OR)

(c) A rectangular box open at the top is to have volume 32 cubic ft. Find the dimensions of the box requiring least material for its construction. 5M

(d) Find the maximum and minimum values of  $f(x,y) = x^3 + y^3 - 3axy$  5M

9. (a) If  $F = (5xy - 6x^2)i + (2y - 4x)j$  Evaluate  $\int_c \vec{F} \cdot d\vec{r}$  where  $c$  is the curve in the  $xy$  plane  $y = x^3$  from  $(1,1)$  to  $(2,8)$  5M

(b) Find  $\text{div } \vec{f}$  when  $\vec{f} = \text{grad} (x^3 + y^3 + z^3 - 3xyz)$  5M

(OR)

(c) Verify Green's theorem in the plane for  $\oint (3x^2 - 8y^2)dx + (4y - 6xy)dy$  where  $C$  is the region bounded by  $y = \sqrt{x}$  and  $y = x^2$  10M

10. (a) Solve the following system of equations using Gauss Seidel iteration method  
 $10x + 2y + z = 9$ ,  $x + 10y - z = -22$ ,  $-2x + 3y + 10z = 22$  10M

(OR)

(c) Using Newton's divided difference formula evaluate  $f(8)$  given that 10M

|      |    |     |     |     |      |      |
|------|----|-----|-----|-----|------|------|
| x    | 4  | 5   | 7   | 10  | 11   | 13   |
| f(x) | 48 | 100 | 294 | 900 | 1210 | 2028 |

**Stanley College of Engineering and Technology for Women(A)****B.E (CSE,CME & AI&DS) I Semester (Backlog) Examinations January-2023****Essence of Indian Traditional Knowledge****Time: 3 hours****Max.Marks:60****PART-A****Note: Answer all questions (Compulsory)****5 X 2=10M**

- |                                             |           |
|---------------------------------------------|-----------|
| 1 What is heritage?                         | 2M L1 CO1 |
| 2 What are Puaranas?                        | 2M L1 CO2 |
| 3 Define religion.                          | 2M L1 CO3 |
| 4 Write a short note on Indian Handicrafts. | 2M L1 CO4 |
| 5 What is value based education?            | 2M L1 CO5 |

**PART-B****Note: Answer all questions****5 X 10= 50M**

- |                                                        |           |
|--------------------------------------------------------|-----------|
| a. What is the importance of culture in human life?    | 5M L1 CO1 |
| b. Give a detail account of Indus Valley Civilization. | 5M L1 CO1 |

**OR**

- |                                                                                      |           |
|--------------------------------------------------------------------------------------|-----------|
| c. Discuss the salient features of Medieval India.                                   | 5M L2 CO1 |
| d. Explain the relationship between culture and heritage.                            | 5M L2 CO1 |
| a. Summarize the developments of South Indian Languages and Literature.              | 5M L2 CO2 |
| b. What is the role of Sanskrit in the evolution of Indian languages and literature? | 5M L1 CO2 |

**OR**

- |                                                                             |           |
|-----------------------------------------------------------------------------|-----------|
| c. Write in detail about the four Vedas.                                    | 5M L2 CO2 |
| d. Discuss the significance of scriptures to the current Indian Society.    | 5M L2 CO2 |
| a. Examine the salient features of Heterodox schools of Indian Philosophy . | 5M L1 CO3 |
| b. Write about Religious Reform Movements in Modern India.                  | 5M L1 CO3 |

**OR**

- |                                                                                |           |
|--------------------------------------------------------------------------------|-----------|
| c. Discuss the relationship between Religion and Philosophy.                   | 5M L2 CO3 |
| d. Write an detailed essay on Bakthi Movement and Sufi Movement.               | 5M L1 CO3 |
| a. Explain in detail about the development of Science and Technology in India. | 5M L2 CO4 |
| b. Discuss about the Indian Architecture of modern India.                      | 5M L2 CO4 |

**OR**

- |                                                                                                     |           |
|-----------------------------------------------------------------------------------------------------|-----------|
| c. Discuss important Indian paintings in detail.                                                    | 5M L2 CO4 |
| d. Elucidate the development of science from ancient period to modern period in India               | 5M L2 CO4 |
| a. Examine the role of education in cultural development.                                           | 5M L2 CO5 |
| b. Write about the modern Indian scientists who contributed to the field of Science and Technology. | 5M L1 CO5 |

**OR**

- |                                                                            |           |
|----------------------------------------------------------------------------|-----------|
| c. Discuss the Education system of ancient India.                          | 5M L2 CO5 |
| d. Explain the important aspects of National Education Policy (NEP) -2020. | 5M L2     |

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**Stanley College of Engineering and Technology for Women(A)****B.E(CSE,CME & AI&DS) I Semester (Backlog) Examinations January-2023****Applied Physics**

Time: 3 hours

Max.Marks:60

**PART-A****Note: Answer all questions (Compulsory)****5 X 2=10M**

- |   |                                                                   |    |        |
|---|-------------------------------------------------------------------|----|--------|
| 1 | List any four applications of optical fibers.                     | 2M | CO1 L3 |
| 2 | Distinguish between soft and hard magnetic materials.             | 2M | CO2 L2 |
| 3 | What is meant by a wave function? Give its physical significance. | 2M | CO3 L2 |
| 4 | What are the limitations classical free electron theory?          | 2M | CO4 L1 |
| 5 | Write a note on surface to volume ratio at nano scale.            | 2M | CO5 L1 |

**PART-B****Note: Answer all questions****5 X 10= 50M**

- |   |                                                                                                                   |    |        |
|---|-------------------------------------------------------------------------------------------------------------------|----|--------|
| 6 | a. With the help of neat sketch elaborate the construction and working of He-Ne laser.                            | 5M | CO1 L3 |
|   | b. Construct the relationship between Einstein's co-efficient of absorption, spontaneous and stimulated emission. | 5M | CO1 L4 |

**OR**

- |  |                                                                                                                                                                                                                            |    |        |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------|
|  | c. Discuss various types of optical fibers with necessary illustrations.                                                                                                                                                   | 5M | CO1 L2 |
|  | d. Calculate (a) numerical aperture (b) acceptance angle and (c) fractional change of index of an optical fiber with following specifications: refractive index of core is 1.36 and refractive index of cladding is 1.326. | 5M | CO1 L3 |

- |   |                                                                                                  |    |        |
|---|--------------------------------------------------------------------------------------------------|----|--------|
| 7 | a. How do you determine dielectric constant of a given material using capacitance bridge method? | 5M | CO2 L3 |
|---|--------------------------------------------------------------------------------------------------|----|--------|

- |  |                                                                             |    |        |
|--|-----------------------------------------------------------------------------|----|--------|
|  | b. Explain ionic polarizability of a dielectric material and give examples. | 5M | CO2 L3 |
|--|-----------------------------------------------------------------------------|----|--------|

**OR**

- |  |                                                                                                                |    |        |
|--|----------------------------------------------------------------------------------------------------------------|----|--------|
|  | c. Describe Weiss molecular field theory of ferro magnetism and hence deduce expression for Curie Temperature. | 5M | CO2 L2 |
|--|----------------------------------------------------------------------------------------------------------------|----|--------|

- |  |                                                                   |    |        |
|--|-------------------------------------------------------------------|----|--------|
|  | d. Draw and Explain Hysteresis curve of a ferromagnetic material. | 5M | CO2 L2 |
|--|-------------------------------------------------------------------|----|--------|

- |   |                                                            |    |        |
|---|------------------------------------------------------------|----|--------|
| 8 | a. Examine various types of superconductors with examples. | 5M | CO3 L1 |
|---|------------------------------------------------------------|----|--------|

- |  |                                                                                                                                                                                |    |        |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------|
|  | b. The critical magnetic field and critical temperature of superconducting lead (Pb) respectively are $6.37 \times 10^4$ A/m and 7.2K. Calculate critical field of lead at 5K. | 5M | CO3 L3 |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------|

**OR**

- c. Analyse using Schrodinger wave equation the energy levels of a particle confined to one dimensional box. 5M CO3 L4
- d. Write a note on de Broglie Matter waves. Calculate wavelength associated with an electron moving with a velocity equal of  $1/15$  of speed of light. 5M CO3 L3
- 9 a. Appraise the classification of materials based on Energy bands in solids with suitable examples. 5M CO4 L4
- b. List the significant conclusions of Kronig-Penney model. 5M CO4 L2
- OR**
- c. Deduce an expression for the carrier concentration in an intrinsic semiconductor. 5M CO4 L2
- d. State Hall effect and mention its applications. 5M CO4 L3
- 10 a. Illustrate working of Electron beam evaporation method to prepare a thin film. 5M CO5 L2
- b. Compare between bulk, thin films and nanomaterials. 5M CO5 L5
- OR**
- c. Elaborate the properties of carbon nanotubes. 5M CO5 L1
- d. Explain the preparation on nanomaterials by sol-gel method. 5M CO5 L2

# Stanley College of Engineering and Technology for Women(A)

B.E (CSE/EEE/ECE/IT/CME/AI&DS) I Semester (Backlog) Examinations January-2023

## Programming for Problem Solving

Time: 3 hours

Max.Marks:60

### PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- 1 Define implicit type casting with example. 2 M
- 2 Differentiate between 'break' and 'continue' statements. 2 M
- 3 Compare call by value and call by reference. 2 M
- 4 What are the bit fields in C? 2 M
- 5 What is the use of fseek( ) function in files. Write its syntax? 2 M

### PART-B

Note: Answer all questions

5 X 10= 50M

- 6 a. Discuss about different computer languages with examples. 5 M
- b. What is an operator? List and explain various types of operators. 5 M

OR

- c. Explain about the basic data types in c language with example. 5 M
- d. Draw the flowchart and write a C program to compute simple interest. 5 M
- 7 a. Explain the for loop and nested for loop with suitable examples. 5 M
- b. Write a C program to find the sum of first and last digit of a number. 5 M

OR

- c. Explain switch statement with syntax and example. 5 M
- d. Implement a C program to find the reverse of an integer number and check whether it is palindrome or not. 5 M
- 8 a. What is an array? What are the different ways of initializing the arrays? 5 M
- b. Explain different string handling functions with syntaxes. 5 M

OR

- c. Explain how arguments are passed to a function using references? 5 M
- d. Demonstrate storage classes available in C language with examples. 5 M
- 9 a. What is Structure? Explain initialization and accessing structures. 5 M
- b. What are enumeration? Give an example to explain the same. 5 M

OR

- c. Explain the concept of Nested structures with a sample C program. 5 M
- d. What is dynamic memory management? Explain different dynamic memory management functions available in C language. 5 M
- 10 a. Explain command line arguments with an example. 5 M
- b. Write a program to perform Linear Search operation. 5 M

OR

- c. Demonstrate file types and file opening modes. 5 M
- d. Explain bubble sort operation with an example 5 M